

Emmanuel Owusu Asante

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SUMMARY

Machine-learning practitioner with an MPS in Analytics (Machine Intelligence) and a mathematics background. Hands-on work across computer vision, clinical predictive modeling, and reproducible data analysis in Python.

EDUCATION

Northeastern University 2025
Master of Professional Studies, Analytics — Concentration: Machine Intelligence

TECHNICAL SKILLS

Languages Python, MATLAB, SQL
ML & CV PyTorch, scikit-learn, CNNs, semantic segmentation, transformers (SegFormer / ViT), OpenCV / scikit-image
Data pandas, NumPy, matplotlib, seaborn, Jupyter, cross-validation, EDA, feature engineering
Tools Git / GitHub, ImageJ / Fiji, PyImageJ, Linux

SELECTED PROJECTS

LightCSPNet — Cloud Detection & Segmentation, Landsat 8 2026
Independent research · Python, PyTorch

- Built a compact CNN encoder–decoder for pixel-level cloud segmentation, progressing from USGS EarthExplorer scenes and Otsu pseudo-masks to the annotated 38-Cloud benchmark.
- Reported IoU 0.73 / Dice 0.85 — ahead of a CLIP patch classifier and zero-shot SegFormer / ViT baselines.

Spinal Abscess Prediction from ICU Records 2024
Northeastern University, EAI 6010 · team of 2 · Python, scikit-learn

- Constructed a case/control cohort from the MIMIC-III critical-care database, joining admissions, diagnoses, procedures, and patient tables.
- Trained and evaluated a logistic-regression classifier with stratified 5-fold cross-validation: AUC 0.776 (95% CI 0.71–0.87), sensitivity 0.69, specificity 0.65.

Particle-Size Analysis with ImageJ & PyImageJ 2024
Research collaboration · Python, PyImageJ / Fiji

- Scripted headless Fiji from Python to batch-segment metal-powder micrographs (Ti-6Al-4V, ball-milled powders) and derive size distributions, with downstream regression comparing measurement methods.

Demand Forecasting & Pricing Analytics — TruRoots 2025
Northeastern University, ALY 6980 Capstone · client-sponsored team project · Python

- Compared ARIMA, LSTM, and XGBoost demand forecasts against seasonal/weather/promotional drivers for a premium juice brand's seasonal SKUs.
- Modeled price elasticity and discount-depth response, and segmented customers by price sensitivity via K-means.

NearPeer Enrollment Analysis 2024
Northeastern University, ALY 6080 · team of 4 · Python, pandas

- Merged platform-engagement data with university records; cleaned, joined, and engineered time-based features for exploratory analysis of engagement vs. enrollment.

MCQ Marking Software

MATLAB · Image Processing Toolbox

- Desktop GUI that auto-grades scanned multiple-choice answer sheets against a marking scheme using thresholding and connected-component analysis.

EXPERIENCE

Curriculum & Autograding — iZen AI Academy

- Author Python, NumPy, pandas, and introductory machine-learning course materials and GitHub Classroom autograders used across multiple cohorts.